

STYLMORPH

Major Project

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Outline

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The proposed platform allows users to create realistic 3D avatars of themselves by entering body measurements like height, weight, and body shape.

Users can virtually try on clothes from various e-commerce platforms using product links. The clothing items are mapped accurately onto the 3D model.

The system also uses AI to recommend matching outfits and color combinations based on the user's profile. Available as both web and mobile application for wide accessibility.

Problem Statement

- Existing e-commerce platforms lack personalized and realistic virtual try-on systems.
- Most existing tools use 2D overlays or generic 3D avatars that do not match users' body measurements, shapes, or skin tones.
- Shoppers cannot accurately visualize how clothing will fit and appear on their own bodies, leading to dissatisfaction.
- Inaccurate previews result in high return rates, reduced purchase confidence, and lower trust in online platforms.

Existing System

- Uses static 2D profiles (age,gender,preferences).
- Relies on browsing purchase history for recommendations.
- Product visualization is limited to 2D images/videos.
- Provides only basic personalization (similar items, trending products)

Mastering 3D Modeling in Blender

Authors: Md Saikat Hosen, Shahed Ahmmed, Sreekanth Dekkati

Key Features

- Versatile interface for 3D modeling, sculpting, and rendering.
- Node-based shading and compositing for realism.
- Animation tools including rigging and inverse kinematics.

Advantages

Free and open-source.
Powerful modeling tools.
Large plugin community.

Disadvantages

Steep learning curve.
High system requirements.
Complex UI for beginners.

AG3D – Learning to Generate 3D Avatars

Authors: Zijian Dong, Xu Chen, Jinlong Yang et al.

Key Features

- Generates 3D avatars from 2D images using Fast-SNARF deformation.
- Tri-plane feature representation for geometry detail.
- Adversarial training for realism.

Advantages

No 3D scans required.
Realistic textures.
Captures loose clothing.

Disadvantages

Needs high-end GPU.
Not real-time.
Complex setup.

CAPE Clothing Model

Authors: Qianli Ma, Jinlong Yang, Anurag Ranjan et al.

Key Features

- Extends SMPL with generative clothing layers.
- Uses VAE-GAN for realistic cloth deformation.
- Pose-dependent offsets for flexibility.

Advantages

Realistic wrinkles.
Compatible with SMPL.
Adapts to poses.

Disadvantages

Requires 3D dataset.
Limited to trained garments.
Expensive training.

AG-UI: Enabling User-Facing AI Agents

Authors: Bhavya Ghai, Shubham Gupta, Anirudh Goyal et al.

Key Features

- Framework for integrating AI agents into user interfaces.
- Focus on explainability and transparency.
- Supports adaptive UI generation based on context.

Advantages

Improves user interaction.
Adaptive and responsive.
Supports modularity.

Disadvantages

High initial design effort.
Complex implementation.
Resource-intensive.

Replacing AI Agents for Backend

Authors: J. Rivera, A. Chavan, NCFT 2025

Key Features

- Automates backend creation with AI agents.
- Uses NLP-based intent recognition for API generation.
- Integrates with no-code frameworks.

Advantages

Speeds up development.
Low code knowledge required.
Auto API generation.

Disadvantages

Limited flexibility.
Security concerns.
Dependency on dataset.

DeepFashion3D Dataset

Authors: X. Zhang, J. Ma, R. Zhang et al.

Key Features

- Large-scale 3D clothing dataset.
- Includes diverse garment types and poses.
- Enables training for realistic cloth deformation.

Advantages

High-quality scans.
Supports ML training.
Wide range of clothing.

Disadvantages

Requires huge storage.
Limited real-world variation.
Not real-time ready.

Physically-Based Cloth Simulation

Authors: M. Baraff, D. Terzopoulos

Key Features

- Physics-driven model for cloth deformation.
- Uses finite element analysis for precision.
- Simulates realistic motion and folds.

Advantages

Highly realistic motion.
Physically accurate.
Applicable in films/games.

Disadvantages

Computationally expensive.
Difficult real-time use.
Needs fine-tuning.

Neural Radiance Fields (NeRF)

Authors: Ben Mildenhall, Pratul Srinivasan et al.

Key Features

- Synthesizes 3D scenes from 2D images.
- Uses volumetric rendering with neural networks.
- Enables photorealistic view synthesis.

Advantages

High visual realism.
Accurate depth estimation.
Reusable models.

Disadvantages

Slow training.
Requires multiple images.
GPU intensive.

ScanAva: Scanning-based Avatar Generation

Authors: Y. Li, X. Chen, H. Wang et al.

Key Features

- Captures 3D human models via scanning.
- Enables measurement-based avatar fitting.
- Provides precise geometry and texture.

Advantages

Accurate body models.
Realistic texture mapping.
Suitable for custom fitting.

Disadvantages

Requires scanner hardware.
Limited scalability.
Expensive process.

Literature Survey Summary(1/4)

No	Title	Methodology	Merits	Demerits
1	Mastering 3D Modeling in Blender	Blender-based modeling, sculpting, rendering with node-based shading, animation and compositing tools	Free, open-source, powerful tools, multi-format export, wide community	Steep learning curve, high system requirements, complex interface, needs plugins
2	AG3D: Learning to Generate 3D Avatars	Generates 3D avatars from 2D images using Fast-SNARF, tri-plane features, adversarial training	No 3D scans, realistic geometry, loose clothing captured, pose control	High GPU demand, not real-time, complex training, over-fitting risk
3	CAPE Clothing Model	Extends SMPL model using VAE-GAN, graph convolutions, pose-dependent garment offsets	Realistic cloth simulation, SMPL compatible, captures wrinkles, adaptable to poses	Needs 3D scans, limited flexibility, expensive training, predefined garments

Literature Survey Summary(2/4)

No	Title	Methodology	Merits	Demerits
4	3D Shoe Reconstruction for AR	3D Gaussian splatting from videos, YOLOv8 + SAM for segmentation, AR deployment	Real-time AR fitting, accurate models, mobile-ready, reduces physical trials	Shoe-specific, manual mesh cleanup, not for complex apparel, device limits
5	3D Product Display's Impact	Mixed-method UX research, 3D vs 2D comparisons, statistical and interview analysis	Better than 2D, multi-angle views, boosts buyer confidence, immersive	Time-consuming models, tech infrastructure required, scalability, compatibility
6	Virtual Fitting Room	3D dummy creation from body measurements using Blender, Three.js, real-time garment mapping	Reduces size/fit issues, personalized, efficient, inclusive	Depends on measurement accuracy, limited templates, sparse data

Literature Survey Summary(3/4)

No	Title	Methodology	Merits	Demerits
7	FashionNet Recommendation System	ResNet50 for feature extraction, kNN outfit match, hybrid transfer learning architecture	Fast, accurate, robust features, scalable recommendations	Cold-start problem, high computation, sparse feedback
8	Try-On Technology Review	Systematic review of 80 papers, 7 tech categories, TAM/UTAUT2 frameworks, purchase intent	Comprehensive insights, adoption factors, roadmap for future	Inconsistent data, cultural gaps, no longitudinal data, real-world inaccuracy
9	3D Model Management	Stakeholder interviews, Sayduck tool analysis, AR integration, web optimization	Product understanding, customization support, return reduction	Initial investment, accessibility, performance impact, maintenance

Literature Survey Summary(4/4)

No	Title	Methodology	Merits	Demerits
10	SIZER: 3D Clothing Size Modeling	SIZER dataset (2000 scans, 100 subjects, 10 garments), ParserNet for automatic garment parsing, SizerNet for size-sensitive deformation, SMPL+G framework	First real clothing size dataset, automatic parsing without segmentation, realistic size deformation, enables virtual try-on	Limited to A-pose, requires high-quality scans, complex pipeline, computational intensive

System Requirements

Hardware Requirements:

- **Server:** 8GB RAM, 500GB Storage, Multi-core Processor
- **Client:** 4GB RAM, WebGL compatible browser
- **Mobile:** Android/iOS with ARCore/ARKit support

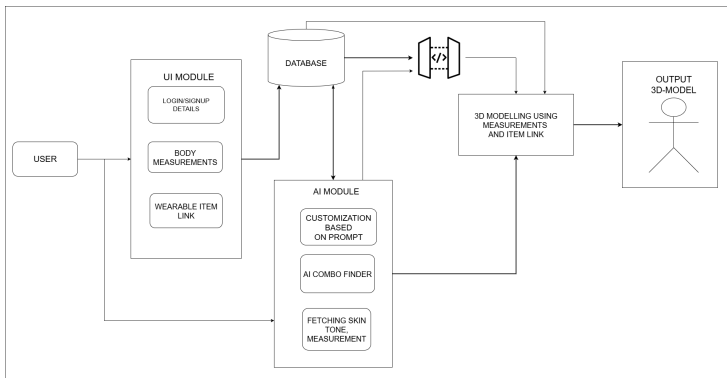
Software Requirements:

- **Frontend:** React.js/Flutter, Three.js for 3D rendering
- **Backend:** Node.js/Python, TensorFlow/PyTorch for AI
- **Database:** MongoDB/PostgreSQL for user data
- **3D Modeling:** Blender, SMPL model integration

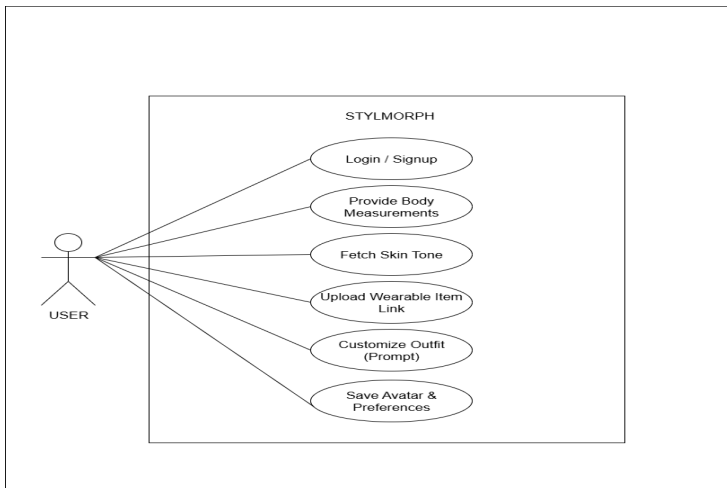
Proposed System

- The system generates a personalized 3D avatar for each user using precise body measurements.
- Users can visualize and try on apparel directly from e-commerce product links in a virtual environment.
- Each user model is uniquely designed to ensure realistic fitting and accurate size representation.
- The system integrates skin tone adaptation to enhance realism and personalization in 3D rendering.
- An intelligent recommendation feature assists users in discovering compatible dress and color combinations.

System Architecture

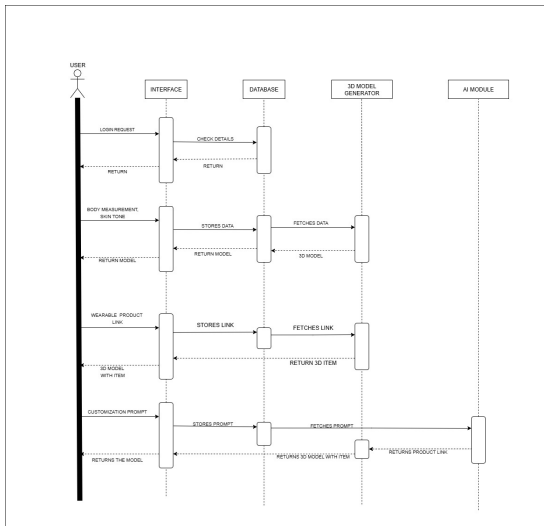


Use Case Diagram

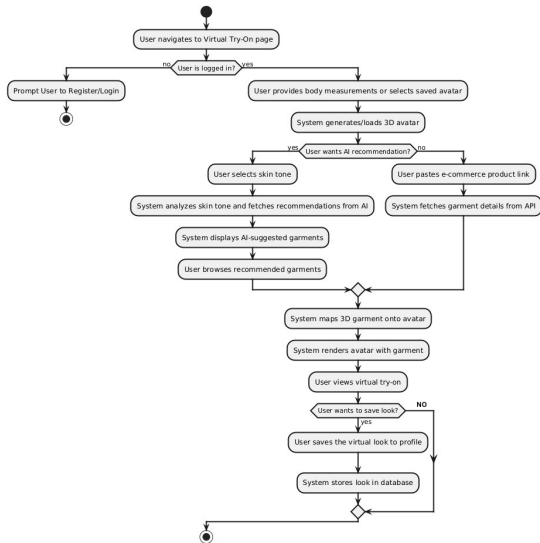


System Design

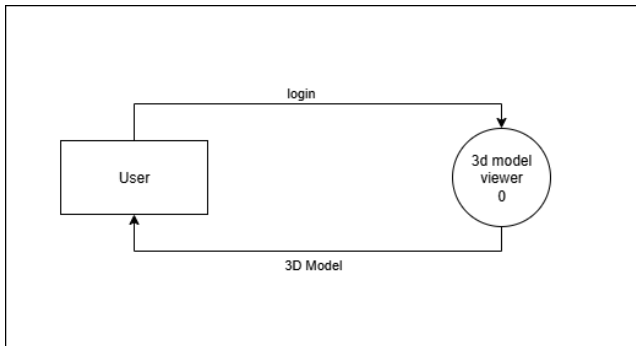
Sequence Diagram



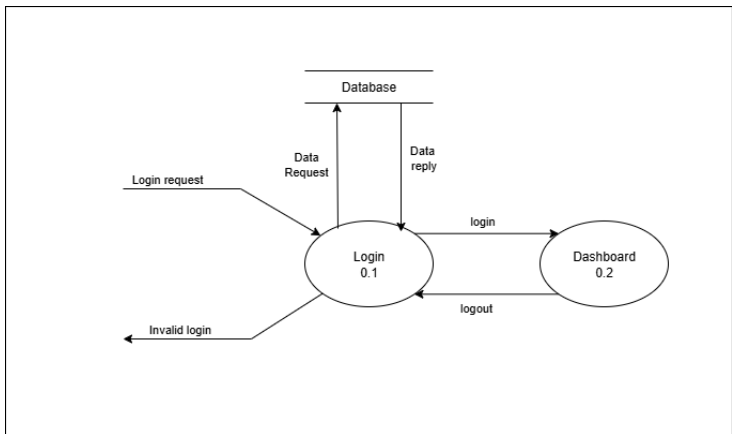
Activity Diagram



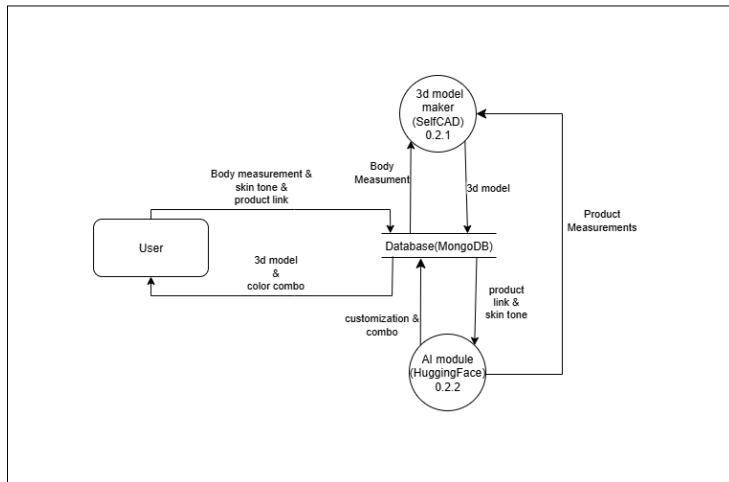
Data Flow Diagram - Level 0



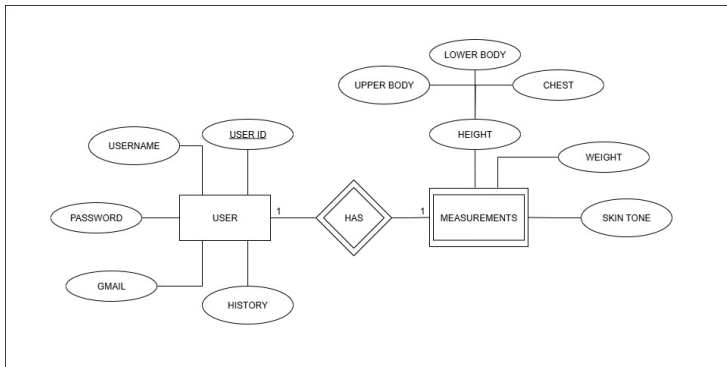
Data Flow Diagram - Level 1



Data Flow Diagram - Level 2

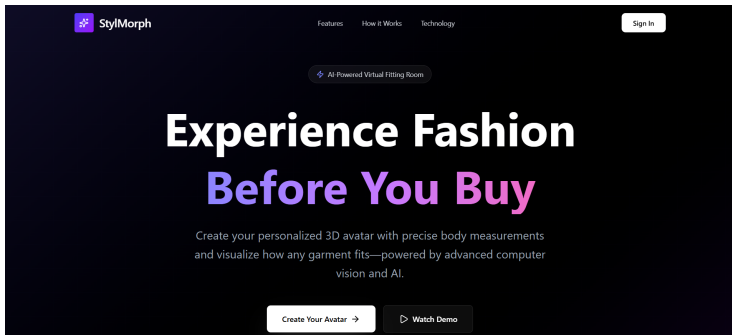


Entity-Relationship Diagram



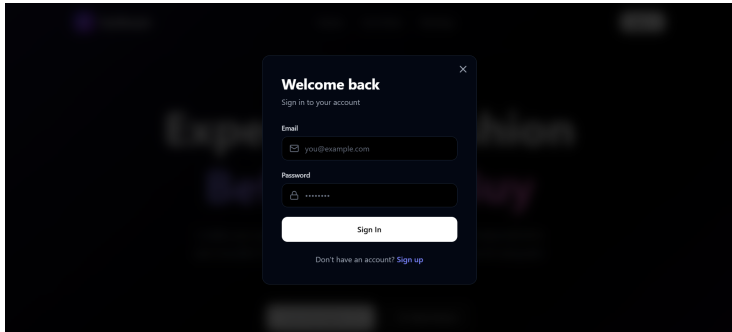
Implementation

Home Screen



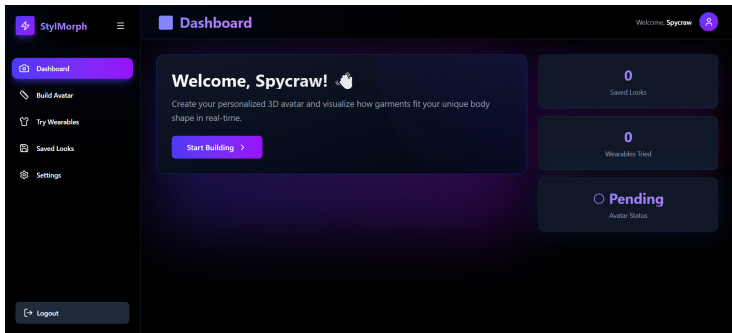
Implementation

Sign Up



Implementation

Dashboard



Implementation

Build Avatar

StylMorph

Build Your Avatar

Welcome, **Spycrow**

Dashboard

Build Avatar

Try Wearables

Saved Looks

Settings

Logout

Enter measurements to see your avatar

Body Measurements

Height (cm)
e.g., 180

Chest (cm)
e.g., 100

Waist (cm)
e.g., 80

Hips (cm)
e.g., 90

Shoulder Width (cm)

Create Avatar

Implementation

Ready Me



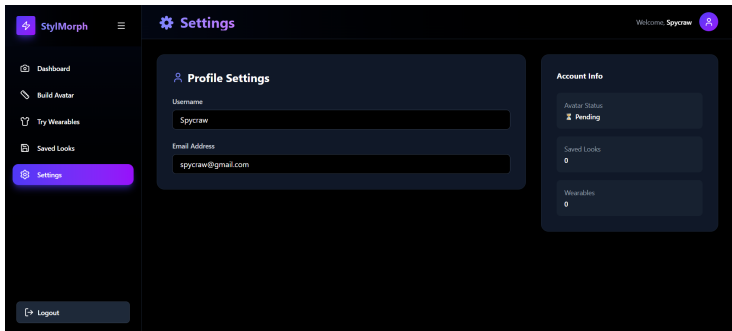
Implementation

Rendered Avatar



Implementation

Settings



- Reduces product returns by helping users choose the right fit and size.
- Enhances shopping experience with personalized, realistic virtual try-ons.
- Matches clothes to skin tone for better style and visual appeal.
- Saves time with smart outfit and color combination suggestions.

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Thank You!